

DYNAMICAL SYSTEMS PROBLEM SET 2

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We begin by noting that all of the files referenced in this document are available at:

<https://github.com/chtompki/MathFall2026/tree/main/DynamicalSystems/ProblemSet2>

The next exercises (3.4.5, 3.5.7, and 3.4.9) are designed to test your ability to distinguish among the various types of bifurcations — its easy to confuse them! In each case, find the values of r at which bifurcations occur, and classify those as: saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork. Finally sketch the bifurcation diagram of fixed points x^* vs r .

3.4.5 $\dot{x} = r - 3x^2$

Solution.

□

3.4.7 $\dot{x} = 5 - re^{-x^2}$

Solution.

□

3.4.9 $\dot{x} = x = \tanh(rx)$

Solution.

□

3.4.11 (An interesting bifurcation diagram) Consider $\dot{x} = rx - \sin(x)$

- a) For the case $r = 0$, find and classify all the fixed points, and sketch the vector field.
- b) Show that when $r > 1$, there is only one fixed point. What kind of fixed point is it?
- c) As r decreases from ∞ to 0, classify *all* the bifurcations that occur.
- d) For $0 < r \ll 1$, find an approximate formula for all values of r at which bifurcations occur.
- e) Now classify all the bifurcations that occur as r decreases from 0 to $-\infty$.
- f) Plot the bifurcation diagram for $-\infty < r < \infty$, and indicate the stability of the various fixed points.

Solution.

□

3.4.14 (Subcritical pitchfork) Consider the system $\dot{x} = rx + x^3 - x^5$, which exhibits a subcritical pitchfork bifurcation.

- a) Find algebraic expressions for all the fixed points as r varies.
- b) Sketch the vector fields as r varies. Be sure to indicate all the fixed points and their stability.
- c) Calculate r_s , the parameter value at which the nonzero fixed points are born in a saddle-node bifurcation.

Solution.

□

3.4.16 (Potentials) In parts (a)-(c) let $V(x)$ be the potential, in the sense that $\dot{x} = -dV/dx$. Sketch the potential as a function of r . Show all qualitatively different cases including bifurcation values of r .

- a) (Saddle-node) $\dot{x} = r - x^2$

- b) (Transcritical) $\dot{x} = rx - x^2$
 c) (Subcritical pitchfork) $\dot{x} = rx + x^3 - x^5$.

Solution.

□

3.5.4 (Bead on a horizontal wire) A bead of mass m is constrained to slide along a straight horizontal wire. A spring of relaxed length L_0 and spring constant k is attached to the mass and to a support point distance h from the wire given in Figure 1 below.

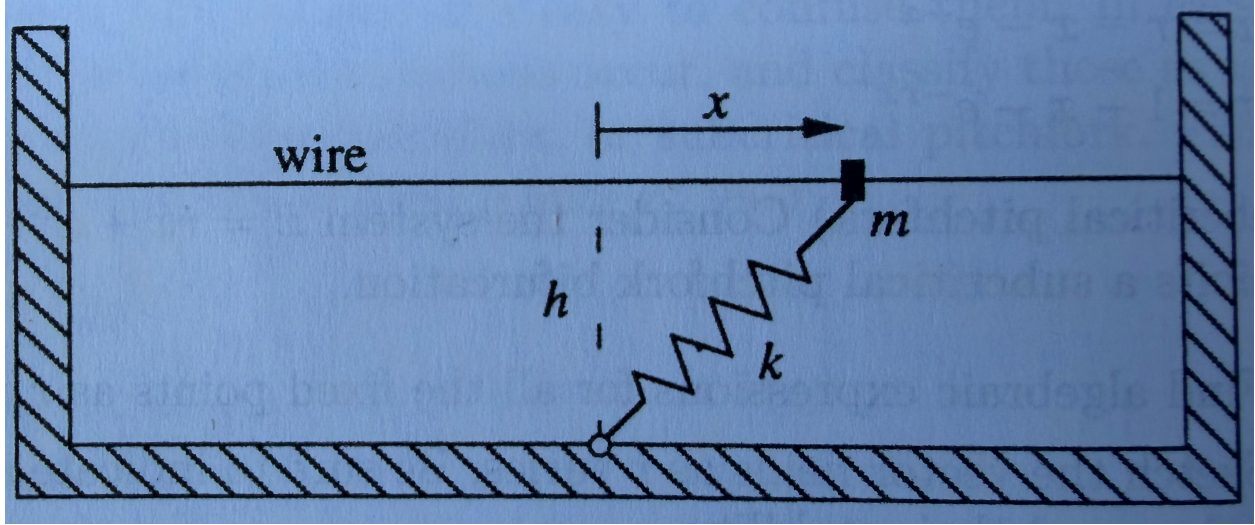


FIGURE 1.

Finally suppose that the motion of the bead is opposed by a viscous damping force $b\dot{x}$.

- Write Newton's law for the motion of the fixed bead.
- Find all possible equilibria, i.e. fixed points, as functions of k , h , m , b , and L_0 .
- Suppose $m = 0$. Classify the stability of all the fixed points, and draw a bifurcation diagram.
- If $m \neq 0$, how small must m be, to be considered negligible? In what sense is it negligible?

Solution.

□

3.6.2 (Imperfect transcritical bifurcation) Consider the system $\dot{x} = h + rx - x^2$. When $h = 0$, this system undergoes a transcritical bifurcation at $r = 0$. Our goal is to see how the bifurcation diagram of x^* vs. r is affected by the imperfection parameter h .

- Plot the bifurcation diagram for $\dot{x} = h + rx - x^2$, for $h < 0$, $h = 0$, and $h > 0$.
- Sketch the regions in the (r, h) plane that correspond to qualitatively different vector fields, and identify the bifurcation that occur on the boundaries of those regions.
- Plot the potential $V(x)$ corresponding to all the different regions in the (r, h) plane.

Solution.

□

3.7.5 (A biochemical switch) Zebra stripes and butterfly wing patterns are two of the most spectacular examples of biological pattern formation. Explaining the development of these patterns is one of the outstanding problems of Biology; see Murray (2003) for an excellent review.

Solution.

□

5.1.10*Solution.***5.2.1***Solution.***5.2.6***Solution.*